

FreeRTOS: A Real-Time Operating System

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Embedded systems have real-time requirements that need to be fulfilled in a given amount of time. This requirement is fulfilled by the real-time operating system (RTOS). In this article we try to understand FreeRTOS. Let us see how this is different from any other RTOS.

FreeRTOS is a part of RTOS that can run even on a small microcontroller (MCU) though is not limited to it. In special cases, like in MCUs, full implementation of an RTOS is not possible because of the size of the MCU and the nature of the end application it is attached to. So FreeRTOS fulfils the requirements for such components in any application.

If you are using an OS on an embedded system, you will have to be very specific about the tasks you want to be done with that embedded system. The OS should be efficient enough to support all those applications, and FreeRTOS allows this.

OS vs FreeRTOS

If FreeRTOS is also an OS, why do we not use it on our computers just like Windows and Linux? Also, why

do we not use Windows or Linux OS on an embedded kit like Galileo or Raspberry Pi, or in an embedded component like an MCU? To understand this, we have to understand both OS and FreeRTOS in detail.

An OS is nothing but a computer

program with the ability to support basic functions of a computer and provide basic services to the applications running on that computer. For example, if you are using Microsoft Office (application that helps you do your file-management work online),

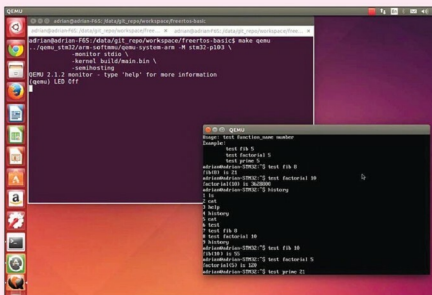


Fig. 1: FreeRTOS shell command using Qemu

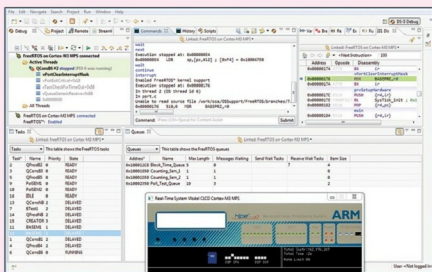


Fig. 2: FreeRTOS and Eclipse views for ARM

Quick review

Version: 8.2.1

Licence: Open source

Applications: Embedded systems

Key features:

- Small size of code
- Independent of third-party source codes

Competing RTOSes

- ChIBIOS
- RTEMS
- NuttX
- uCOS